

# Charge! A New Battle Cry for Energy Efficiency

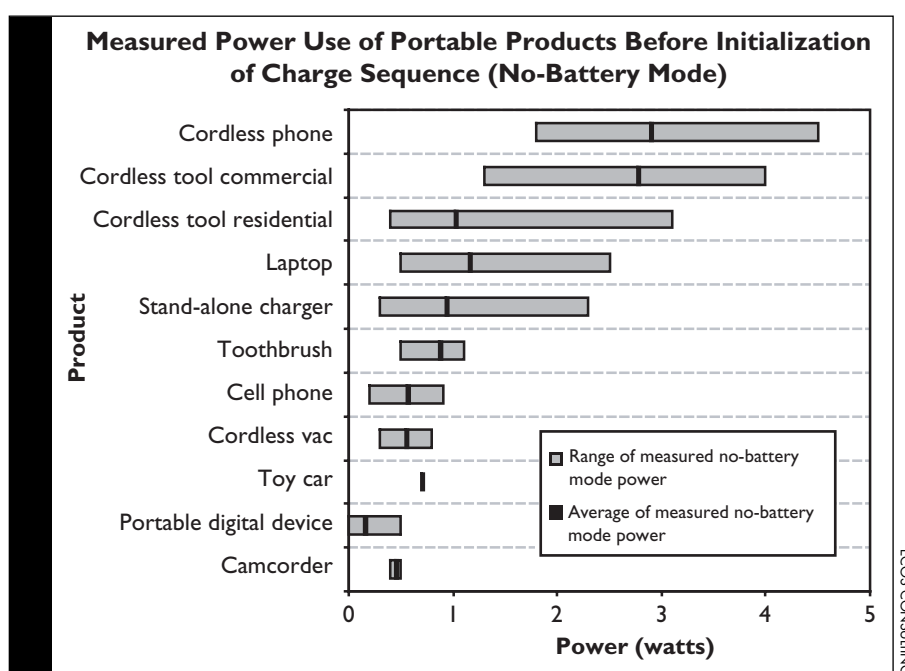
*Beware the rarely used, but still plugged-in, cell phone charger! Rechargeable electronic products are consuming more power than one might think, even when the product is disconnected from the charger.*

by Suzanne Foster  
and Chris Calwell

Over the last two decades, a slew of new portable electronic products have been introduced into the consumer marketplace and are growing steadily in popularity. With the rising sales of camcorders, digital cameras, laptop computers, cordless and cellular phones, PDAs, and cordless appliances and tools, many of us find our homes looking more like a neighborhood Best Buy store than a photo from *Architectural Digest*.

Today, over 600 million products that contain battery chargers are currently in use in America's homes and offices. These products offer substantial economic and environmental advantages over products powered by throwaway batteries and are more convenient than corded products (see "Throwaways Versus Rechargeables"). However, each rechargeable product contains an AC/DC power supply, battery-charging circuitry, and low-voltage electronics whose total electricity consumption varies widely from product to product. To further complicate comparisons, each charger itself consumes varying amounts of electricity, depending on what mode it is in. It may be plugged in but providing no functionality because no device is currently hooked up to it; it may be simply maintaining the charge of an already fully charged battery; or it may be in active charging mode. We call these three operating modes no-battery, battery maintenance, and active charge, respectively.

Over the last two years, Ecos Consulting has been researching these technologies and related efficiency issues for the



**Figure 1.** In no-battery mode, the charger is waiting to accept the battery for charge and in most cases is performing little useful function. Even so, we see power levels as high as 4.2 watts in this mode.

Natural Resources Defense Council (NRDC), the California Energy Commission, and Pacific Gas and Electric Company (PG&E). Through our research, we have attempted to answer the following three questions:

How does low-power mode use vary for different rechargeable products?

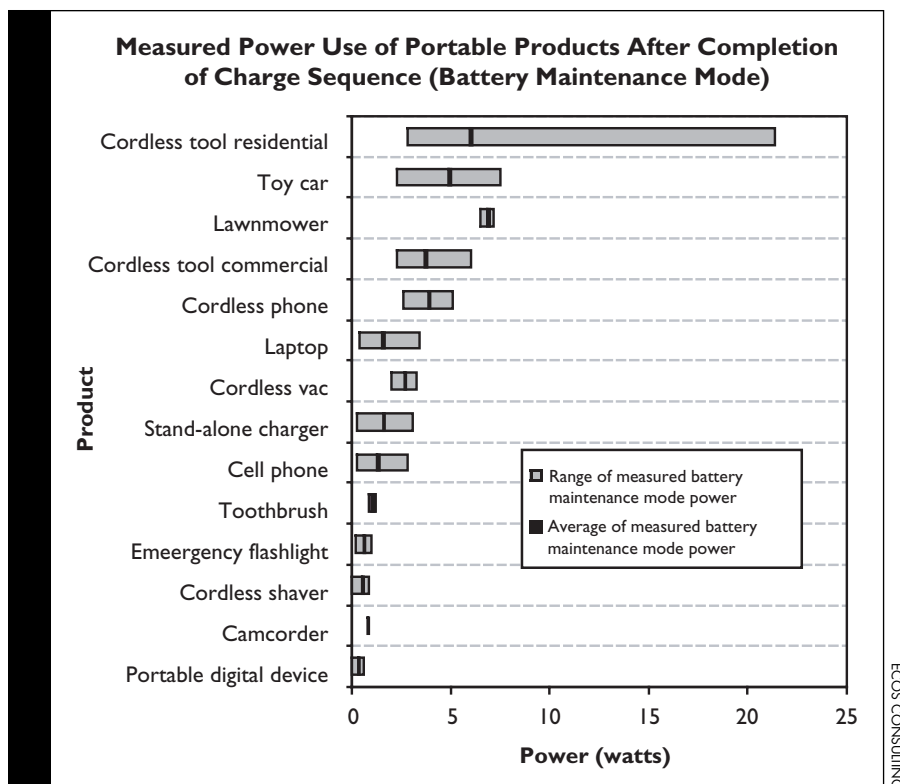
How significant are the differences in charging efficiency?

What can consumers do to maximize the efficiency of rechargeable products?

The most obvious place to begin is with the energy consumed by a charger when it is in no-battery mode, as when a cell phone charger is plugged into the

wall but no cell phone is connected to the charger. Here, power should be as close to zero as possible, yet our measurements show levels of up to 4.2 watts for some cordless phones, with most values falling between 0.5 and 3 watts (see Figure 1).

Next, we considered the case of a charger with a fully charged battery installed. This charger is in battery maintenance mode. Here, power consumption should be near zero much of the time, except for episodic pulses of charging to maintain batteries at full charge and some low-level functionality (like a cordless phone waiting for an



**Figure 2.** The power drawn by a number of chargers after the charge sequence is complete was measured in Ecos Consulting's laboratory. Notice that some portable digital devices draw almost no power in this state. If the technology that draws the least amount of power was incorporated into all similar products, a substantial amount of energy could be saved.

incoming call). Although typical values in this mode are from 2 to 5 watts, antiquated power supply and charger designs often cause rechargeable products to draw anywhere from 5 to more than 20 watts even after their batteries are fully charged (see Figure 2).

Finally, we considered the active charge mode. This is the most complex of the three operating conditions. Power draw is the highest during active

charge mode, so the power supply losses—and the opportunity for savings through improved efficiency—are typically highest in that mode. Yet most consumer battery chargers still employ linear power supplies, often with efficiencies of only 30%–50% (see “How to Measure a Battery Charger”).

At the same time, different battery chemistries have their own inherent efficiency differences associated with the

electrochemical process of storing a charge (see “How Can You Save Energy with Rechargeables?” p. 21). Lithium ion (Li-ion) and nickel metal hydride (NiMH) chemistries tend to be somewhat more efficient than the more common nickel cadmium (NiCad) chemistry. They also yield greater energy densities and longer run times.

Likewise, an individual set of batteries will charge with varying efficiency when placed in different chargers, even if the chargers' power supplies are identically efficient. Universal chargers capable of accepting a wide variety of battery voltages and chemistries can also be less efficient than the dedicated, battery-specific chargers found in cellular phones or laptop computers, for example.

### Active Charge Mode Efficiency

We defined active charge mode efficiency as the DC kWh extractable from the battery under standard discharge conditions divided by the AC kWh required to recharge the depleted battery. This method captures both the losses associated with the power supply's conversion processes and the electrochemical (coulombic) losses associated with charging and discharging the battery. Indeed, these appear to be the most important places to look for energy savings, as the DC motors and electronics in the portable devices are often designed to be fairly efficient in the interest of minimizing physical product size and maximizing battery life.

The best battery charger designs we measured consume only about 25% more AC energy than the amount of

## Throwaways Versus Rechargeables

About 3 billion throwaway batteries are purchased and discarded in the United States each year. Sales of rechargeable batteries are much smaller, but they are growing rapidly; they now total about 300 million units per year. The potential environmental and economic benefits of rechargeable batteries are enormous, with typical

designs lasting for 500 to 1,000 charges before replacement. Even including the greater cost of the batteries, their smaller capacity (compared to most throwaways), the cost of the chargers, and the cost of the electricity to operate them, rechargeable batteries can save consumers hundreds of dollars over the life of the battery compared to

purchasing throwaways. Paybacks are very rapid (typically six months to a few years), depending on how heavily the batteries are used. According to Consumer Reports, using rechargeable NiMH batteries instead of throwaway alkalines in a digital camera saves \$2 for every 100 photos you take.

DC energy that can be extracted from the charged batteries, delivering an efficiency of 75%. The least efficient battery charger designs deliver efficiencies of only 6%, suggesting that portability can come at a heavy price.

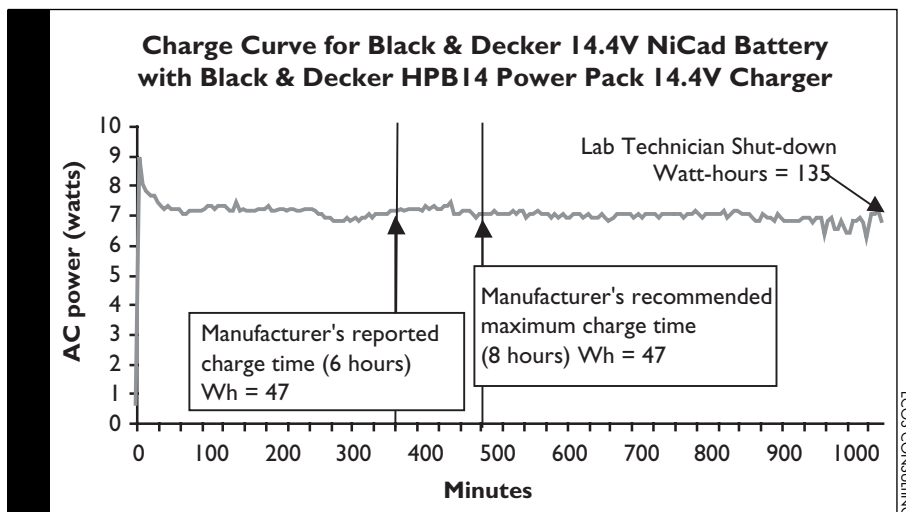
Our measurements of the efficiency of stand-alone AA battery chargers revealed efficiencies as high as 35% and as low as 6%. Much of this has to do with the specific design of the charging system and the chemistry of the battery. The culprit in most cases was the very low efficiency of the external power supplies, along with constant current charger designs that shunt available power to an internal resistor as a battery nears full charge, rather than reducing total DC output power.

We also discovered that charge efficiencies vary even in high-tech products. Laptop computers typically revealed charging efficiencies of 75%, consistent with the use of a highly efficient switching power supply and a precise smart-charging system, where the battery actively informs the charger of its state of charge and temperature to regulate the charging process. On the other end of the spectrum, our testing of three different cellular phones with inefficient power supplies revealed charging efficiencies of about 20%, though simply replacing those power supplies with efficient models could roughly double these phones' efficiencies.

Charging batteries rapidly is attractive to consumers and is not definitively more or less efficient than slow charging. In the extreme case of professional-grade cordless tools and 15-minute stand-alone chargers, ventilation fans are built into the

chargers to enable rapid charging without damaging the batteries. These fans add to the energy consumed by the chargers above and beyond the fundamental chemical efficiency penalties associated with charging a warm battery, which

process, we charged the same cordless drill battery with two different charger designs. The first charger did not have an indicator light and did not power down to a low-power state after the charge was complete (see Figure 3). The



**Figure 3.** The power drawn by this charger after the battery is full does not change. The charger continues to consume energy at the same rate, even though there is no useful work being performed.

accepts charge less easily than a cool battery. On the other hand, a fast charger must be capable of monitoring the battery to ensure that it is not damaging the battery during rapid charge. This monitoring process allows the charger to drop down to a lower power level as soon as the battery is fully charged, thereby saving energy over chargers that pull the same amount of power regardless of the state of charge of the battery.

To nail down how charger design affects the efficiency of the charging

second charger was capable of monitoring the battery and dropping to a low-power state when the battery was full (see Figure 4).

## Low Power, Long Hours

Other investigations suggest that battery chargers consume most of their energy while in no-battery or battery maintenance mode. Arshad Mansoor of EPRI-PEAC recently measured the energy consumption of a BlackBerry

## How to Measure a Battery Charger

**Low-Power Modes.** Plug the battery charger into the power meter with the appropriate configuration (for no-battery mode, no batteries inserted; for battery maintenance mode, fully charged batteries inserted). Wait for the battery charger to cycle to its lowest stable power level and record the power (watts).

**Active Charge Mode.** Plug the battery charger into a power meter that is capable of measuring accumulated energy and insert a fully discharged battery into the charger.

Make sure that all of the battery charger's other functions—the ones not associated with charging the battery—are switched off. Note the meter's starting energy value (watt-hours). Allow the battery to charge until the charger indicates that the battery is full. If the charger is not capable of indicating this, allow the battery to charge for the manufacturer's recommended charge time. Note the energy value on the meter at the end of charge. The difference between the start energy and the end energy is the total

energy consumed by the charger in active charge mode.

If you want to know the efficiency of the charger in active charge mode, use a battery analyzer to fully discharge the battery. The analyzer used should be capable of measuring the energy drawn from the battery while the battery is discharging at a constant current. The total DC energy drawn from the battery divided by the AC energy to charge the battery is the efficiency of the active charge mode.

## How Can You Save Energy with Rechargeables?

Use rechargeable batteries instead of disposable batteries, especially in products that require intensive battery use, like digital cameras and portable CD players.

Purchase chargers with an end-of-charge indicator light so that you can choose to unplug the charger or remove batteries at the end of the charge cycle.

When possible, unplug battery chargers or switch them off with a power strip or an equivalent type of on-off outlet switch after charging is complete. NiCad batteries self-discharge about 10% of their capacity in the first 24 hours and about 20% of their capacity in a month. NiMH batteries, by contrast, self-discharge about

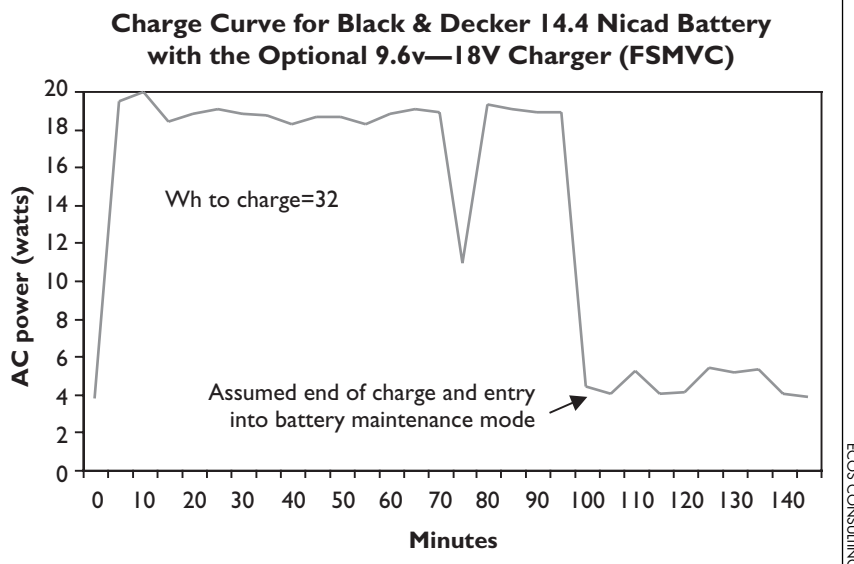
30% of their capacity in a month, but they have a higher overall capacity in a given weight. Better to top them off occasionally than to leave them charging constantly.

Both Nicad and NiMH batteries make poor choices for products that are used very occasionally, such as emergency flashlights. Consider rechargeable alkalines instead of throwaway alkalines. They may only recharge 20 to 40 times, but using them will still save money and resources compared to buying and discarding more throwaways.

At the end of their lifetime, don't forget to take your rechargeables for free recycling to a retailer that works with the Recharge-

able Battery Recycling Coalition (RBRC). NiCad, NiMH, and Li-ion batteries contain a range of different chemicals with varying toxicities, so keeping them out of landfills helps prevent pollution and saves the energy required to manufacture new batteries. See [www.rbrc.org](http://www.rbrc.org) for more information.

Make some measurements; track your usage! While good data exist on energy consumption in various modes, the energy efficiency community still knows very little about hours of use and duty cycles. We need to know how much time battery chargers are spending in their various modes to make better decisions about efficiency opportunities.



**Figure 4.** This charger alerts the user that the battery is full with an indicator light and draws less power once the battery is fully charged. Notice that there is 30% less energy drawn during the charge time for this charger, compared to the 14.4V charger, even though its power consumption, when charging, is significantly higher.

PDA charger (see Figure 5). The charger uses about 2 watts with no battery in the charger, rises to about 6 watts when charging begins, and then steadily declines back to approximately 2 watts when charging is complete. Given the charging time, this suggests that about 3.6 watt-hours (Wh) are required per charge cycle. So a user who leaves the charging base plugged in constantly and charges the PDA once per day would consume about 46 Wh (2 watts x 23 hours) for no-battery and battery maintenance modes combined, and about 3.6

Wh for charging. This means that more than 90% of the energy that is consumed by this battery charger annually is consumed when the charger is plugged in, waiting to accept the PDA for charge.

The situation is similar for many products that we have measured, but without knowing exactly how more complicated products are used, it is difficult to estimate charge mode savings potential accurately. For most rechargeable consumer products, which need fairly infrequent recharging, we are convinced that the total energy consumed in the low power

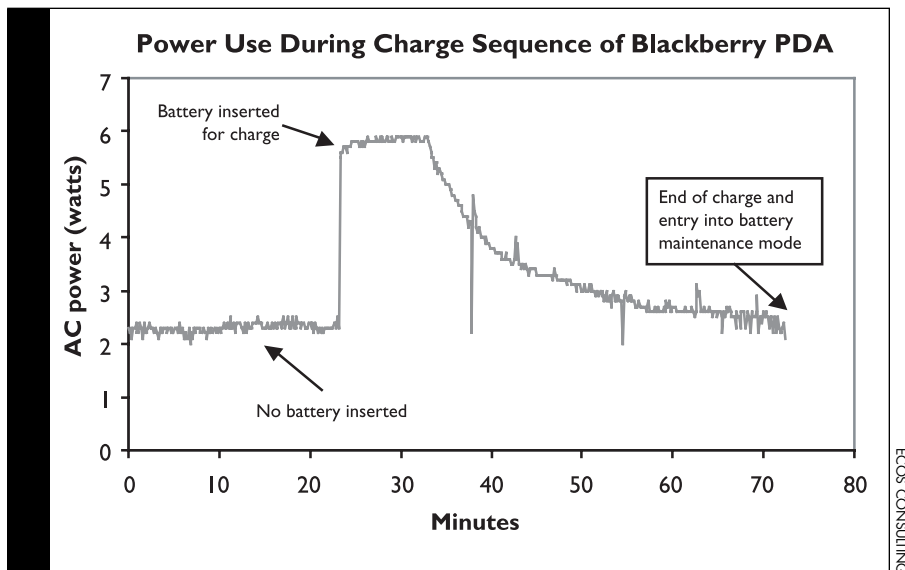
modes makes up most of the total. For products undergoing heavy use in commercial settings, such as a cordless drill for a contractor or AA batteries in a flash or digital camera for a photographer, active charge mode efficiency is likely to have a bigger impact on total energy use.

### Next Steps

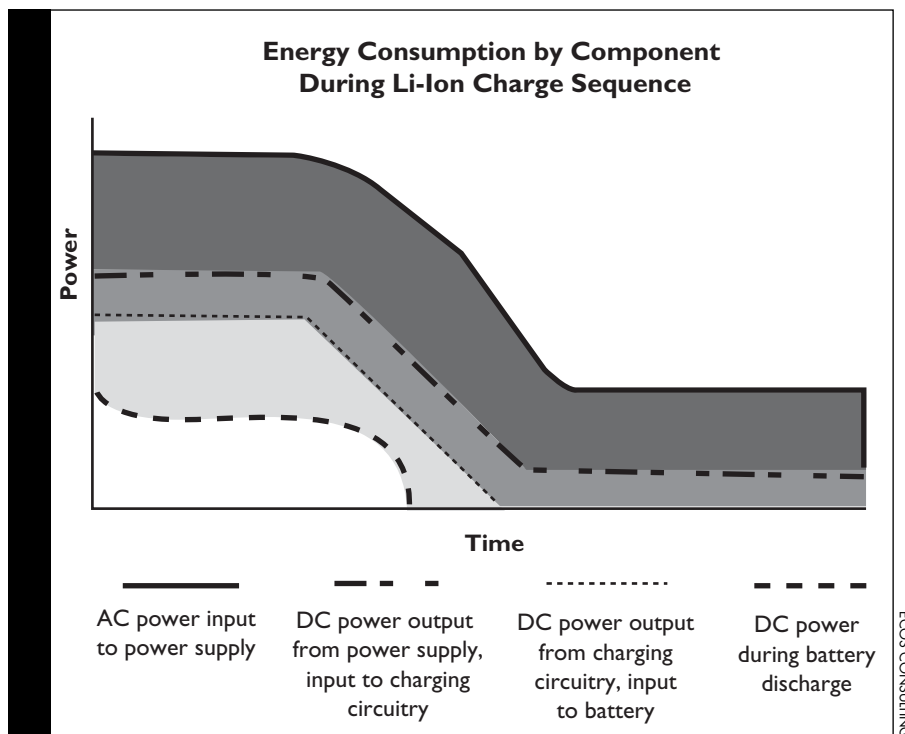
How much low-power mode energy is at stake? We estimate that if all U.S. battery chargers were as efficient in the no-battery and battery maintenance modes as the best designs we tested, the national savings would be about 4.5 billion kWh per year and nearly \$400 million annually.

Energy Star has successfully reduced low-power mode energy consumption in a variety of products, including computers, monitors, and televisions. Battery chargers offer a similar, but new, opportunity for government agencies and utilities. As these organizations contemplate steps they might take in response to this information, we encourage an initial focus on improving power supply efficiency, which will reduce energy consumed in all modes of operation. Next, it makes sense to encourage reductions in no-battery and battery maintenance mode energy consumption in the following ways:

Manufacturers should develop smarter charger designs that drop to little or no power when in no-battery or battery maintenance mode. (An occasional, instead of continuous, topping charge



**Figure 5.** Notice that the PDA requires 2 watts when it is plugged into the wall with no battery. If this unit is plugged in during the 23 hours per day that it is not being used for the charger, this adds up to 46 Wh during a 24-hour period. This is compared to the 3.6 Wh used to charge the battery.



**Figure 6.** The first three shaded regions (top to bottom) give the amount of energy consumed during the charging process by the power supply, battery charging circuitry, and battery, respectively. The area under the fourth curve is the energy that was extractable from the battery during a discharge process. In future research, this type of measurement could help define the important technical aspects of battery charger efficiency.

could ensure that battery maintenance functions are performed.)

Chargers should provide a clear and consistent indication of the fully charged battery condition, so that users can choose to unplug chargers when

they are finished charging. Virtually all charger designs can save energy by providing this type of indicator, since chargers typically consume significantly less power in no-battery mode than in battery maintenance mode (and no

power at all when they are unplugged!).

There should be a hard off switch on the battery charging base to give users a way of eliminating energy use when the device is plugged in but not being used.

More research is needed to further understand the active mode of battery chargers. By measuring the power at different circuitry junctions in the battery charger, it is possible to identify the sources of energy loss throughout the charging process. Figure 6 shows a hypothetical 8-hour period where a Li-Ion battery is charged and then continues to be plugged into the AC wall outlet. The area under each curve (top to bottom) gives the amount of energy consumed by the power supply, by the charger, and by the battery during the charge and standby process.

The lowest dotted curve superimposed on the charge sequence illustrates the battery *discharge* curve in time. The white area underneath this curve is the energy extracted from the battery. Comparing extractable energy to the total energy consumed by the charging process (the area under the top curve) gives the overall efficiency of the battery charger. Notice also that after the battery becomes fully charged, more of the total energy being consumed is being consumed by the power supply itself, even though there is no charging functionality being delivered.

Active charge mode efficiencies are important, especially in products that are frequently charged. Our measurements of active charge mode efficiencies show a wide variation among different products. This variation may be reason enough to consider adopting specifications encouraging the most efficient charge circuitry designs.



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## For more information:

Visit [www.efficientpowersupplies.org](http://www.efficientpowersupplies.org) to learn more about how power supply efficiency affects battery chargers.